METRIC
MIL-R-24737(NAVY)
31 August 1989

MILITARY SPECIFICATION

RECEIVERS, LIGHT SIGNAL, ANALOG, FIBER OPTIC, SHIPBOARD (METRIC), GENERAL SPECIFICATION FOR

This specification is approved for use by the Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for fiber optic analog receivers for use in Naval shipboard systems. The fiber optic analog receiver accepts an analog signal through the input optical interface and converts the optical signal to an electrical output signal. The receiver amplifies and conditions the electrical signal to be compatible with the specified interface. The receiver does not alter the input signal modulation format.

1.2 Classification. Receivers are classified as specified in 1.2.1 through 1.2.3 (see 3.1).

1.2.1 Wavelength class. The wavelength class designation indicates the nominal wavelength of the input optical signals.

(a) Class A - 1.31 micrometers (μm).
(b) Class B - 1.55 μm.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.2 Frequency range (see 6.6.7). The maximum frequency of operation of the receiver falls within one of the following frequency ranges.

(a) Range 1 - 0 to 20 megahertz (MHz).
(b) Range 2 - 20 to 500 MHz.
(c) Range 3 - 500 MHz to 2 gigahertz (GHz).
(d) Range 4 - Greater than 2 GHz.

1.2.3 Style. The style designation defines the style of package.

(a) Style A - Dual inline package using pigtail.
(b) Style B - Dual inline package, connectorized.
(c) Style C - Surface mount package using pigtail.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-T-24735	- Transmitters, Light Signal, Analog, Fiber Optic. Shipboard (Metric) General Specification for.
MIL-T-24735/1	- Transmitter, Analog, Fiber Optic, Shipboard, 0.5 to 60 MHz (0.5 dB Passband) (Metric).
MIL-R-24737/1	- Receiver, Light Signal, Analog, Fiber Optic, Shipboard, 0.5 to 60 MHz (0.5 dB Passband) (Metric).
MIL-M-38510	- Microcircuits, General Specification for.
MIL-M-55565	- Microcircuits, Packaging of.
MIL-C-85045	- Cable, Fiber Optic, Shipboard (Metric) General Specification for.
MIL-C-85045/14	- Cable, Fiber Optic, Fiber Optic Cable Configuration Type B (Pigtail), Optical Fiber Type MM (Graded Index, Glass Core and Glass Cladding, Multimode) (Metric).

STANDARDS

MILITARY

MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.

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MILITARY (Continued)

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines.
MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

455-6 - Cable Retention Test Procedure for Fiber Optic Cable Interconnecting Devices. (DoD adopted)
455-36 - Twist Test for Fiber Optic Connecting Devices. (DoD adopted)

(Application for copies should be addressed to Electronic Industries Association, 2001 Eye Street, NW, Washington, DC 20006.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.3 Reliability. Receiver reliability shall be expressed as mean time between failure (MTBF), in hours of operation, and shall be as specified (see 3.1). Reliability shall be determined as specified (see 4.6.1 and 6.2). Receiver failure criteria shall be as specified (see 3.1).

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3.3.1 Burn-in (see 4.3.2). Burn-in requirements of MIL-STD-883, method 1015 shall be met.

3.4 Materials. Materials shall not have adverse effects upon operational or maintenance personnel under all operational and environmental conditions, nor cause degradation of equipment performance (see 6.3).

3.4.1 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using raw materials produced from recovered bulk materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of partially processed, assembled, or rebuilt products is allowed under this specification.

3.5 Design and construction. The receiver shall be of the design and construction as specified herein.

3.5.1 Fiber optic pigtail. Receiver styles A and C shall be supplied with an optical fiber or optical cable pigtail (see 6.6.11 and 6.6.12) as specified (see 3.1). One end of the pigtail shall be permanently mounted in the receiver case, and the mounting construction may provide strain relief for the pigtail in order to meet the requirements of 3.11. The pigtail length shall be not less than 1 meter. No metal components shall be used within the pigtail, except for mounting to the receiver case or termination connector.

3.5.2 Fiber optic connector. Receiver style B shall be supplied with a fiber optic connector as specified (see 3.1).

3.5.3 Case (see 4.6.2.1). The receiver shall be hermetically sealed. Dimensions, pinouts, and pinout functions shall be as specified (see 3.1). The number of pins on each receiver shall be one of the following: 14, 16, 18, 20, 24, or as specified (see 3.1). The receiver case may be electrically grounded, connected to a voltage pin, or electrically isolated (that is, floating) as specified (see 3.1).

3.5.4 Terminals. The terminals shall meet the requirements as specified in 3.5.4.1 and 3.5.4.2.

3.5.4.1 Terminal strength (see 4.6.2.2). The electrical terminals shall reveal no breakage, loosening, or relative motion between the electrical terminals and the case of the receiver.

3.5.4.2 Solderability (see 4.6.2.3). Terminal solderability shall meet the temperature and time requirements as specified in MIL-STD-883, method 2003, or as specified (see 3.1).

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3.6 Maintainability. The receiver shall require no preventive maintenance.

3.7 Transportability. The receiver operational performance specified herein shall not be affected following prolonged storage or shipment under environmental conditions for shipment and storage specified herein, and as required for packaging for shipment (see 5.1).

3.8 Performance requirements. The receiver shall meet all performance requirements of 3.8.1 and 3.8.2 specified herein.

3.8.1 Optical performance properties.

3.8.1.1 Spectral responsivity (see 4.6.3.2.1). The receiver shall support photodetection over the spectral wavelength range as specified (see 3.1).

3.8.1.2 Dynamic range (see 4.6.3.2.2 and 6.6.4). The receiver shall exhibit a dynamic range at the signal-to-noise ratio (SNR) as specified (see 3.1), without degradation in performance as specified in 3.8.2.

3.8.1.3 Sensitivity (see 4.6.3.2.3 and 6.6.14). The receiver sensitivity shall be as specified (see 3.1) at the specified SNR (see 3.1).

3.8.2 Optical-electrical properties.

3.8.2.1 Signal-to-noise ratio (SNR) (see 4.6.3.3.1 and 6.6.15). The minimum SNR shall be as specified (see 3.1), and shall be maintained over the specified frequency range.

3.8.2.2 Frequency range (see 4.6.3.3.2 and 6.6.7). The receiver shall convert the optical input signal into an electrical signal over the frequency range specified (see 3.1).

3.8.2.3 Total harmonic distortion (THD) (see 4.6.3.3.3 and 6.6.16). The maximum allowable total harmonic distortion shall be as specified (see 3.1).

3.8.2.4 Intermodulation distortion (IMD) (see 4.6.3.3.4 and 6.6.10). The maximum allowable IMD shall be as specified (see 3.1).

3.8.2.5 Amplitude distortion (see 4.6.3.3.5 and 6.6.1). The maximum allowable amplitude distortion shall be as specified (see 3.1).

3.8.2.6 Group delay (see 4.6.3.3.6 and 6.6.8). The maximum allowable group delay shall be as specified (see 3.1).

3.8.3 Electrical properties.

3.8.3.1 Analog output signal (see 4.6.3.4.1 and 6.6.2). The electrical output signal interface requirements for impedance, signal levels, and modulation format shall be as specified (see 3.1).

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3.8.3.2 Short circuit protection (see 4.6.3.4.2). The receiver shall not experience any degradation of performance following application and removal of an electrical short between any required functional pin and logic ground or between any functional pin and supply voltage.

3.8.4 Power supply voltage and current (see 4.6.3.5). The receiver shall operate with one or more of the following nominal direct current (dc) supply voltages: plus and/or minus 5 volts dc, plus and/or minus 12 volts dc, plus and/or minus 18 volts dc, plus and/or minus 24 volts dc, or plus and/or minus 28 volts dc as specified (see 3.1). The receiver shall meet all electrical and optical performance requirements specified herein while operating with specified power supply voltages, currents, and their tolerances. The total receiver power consumption shall not exceed the level as specified (see 3.1).

3.9 Safety. The handling, installation, and operation of the receiver shall not endanger personnel by direct contact or by long-term effects due to deterioration of any component within the receiver construction. Receiver design and operation shall be in accordance with the safety standards of MIL-STD-454, requirement 1.

3.10 Environmental requirements. The receiver shall meet all performance requirements under the following environmental conditions and as specified (see 3.1).

3.10.1 Temperature extremes (see 4.6.4.1). The receiver shall meet the performance requirements of 3.8.1 and 3.8.2 at the operating temperature extremes and there shall be no physical damage (see 4.6.2.4). The receiver temperature range shall be as specified (see 3.1), in accordance with the ranges in table I.

TABLE I. Temperature ranges.

Temperature range	Operating (°C)	Nonoperating (°C)
1	0 to +85	-62 to +85
2	-28 to +85	-62 to +85
3	-54 to +85	-62 to +85
4	0 to +125	-62 to +125
5	-28 to +125	-62 to +125
6	-54 to +125	-62 to +125

3.10.2 Thermal shock (see 4.6.4.2). Over the nonoperating temperature range specified (see 3.10.1), the receiver shall meet the performance requirements of 3.8.1 and 3.8.2, and there shall be no cracks, breaks, or other physical damage (see 4.6.2.4). The receiver shall remain hermetically sealed (see 4.6.2.1).

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3.10.3 Impact shock (4.6.4.3). The receiver shall meet the performance requirements of 3.8.2, and there shall be no cracks, chips, warping, bending, or other degradation (see 4.6.2.4). The receiver shall remain hermetically sealed (see 4.6.2.1).

3.10.4 Vibration (see 4.6.4.4). The receiver shall meet the requirements of 3.8.2, and there shall be no cracks, chips, warping, bending, or other degradation (see 4.6.2.4). The receiver shall remain hermetically sealed (see 4.6.2.1).

3.10.5 Salt spray (see 4.6.4.5). The receiver percent corrosion shall be in accordance with MIL-STD-883, method 1009, test condition C.

3.10.6 Explosive atmosphere (see 4.6.4.6). The receiver shall safely operate in a flammable atmosphere without causing an explosion.

3.10.7 Fungus resistance (see 4.6.4.7). The receiver shall meet the requirements of MIL-STD-454, requirement 4. Materials not identified in MIL-STD-454, requirement 4 for fungus inert materials, shall be in accordance with grade I of MIL-STD-810, method 508.

3.10.8 Environmental screening. The receiver shall be environmentally screened by random vibration (see 4.3.3) and by temperature cycling (see 4.3.4).

3.11 Mechanical requirements (see 4.6.5).

3.11.1 Cable pull-out force (see 4.6.5.1). The minimum cable-to-receiver pull-out strength shall be a minimum of 50 percent of the specified tensile strength of the attached cable or 100 newtons, whichever is less. There shall be no cable jacket damage, cable clamp failure, cable-to-receiver seal damage, distortion or bending of receiver parts, or cable disengagement from the clamp (see 4.6.2.4). The receiver shall meet the requirements specified in 3.8.2.1.

3.11.2 Twist (see 4.6.5.2). Cable-to-receiver seals shall not be damaged nor shall any other physical damage occur (see 4.6.2.4). The receiver shall meet the requirements specified in 3.8.2.1.

3.11.3 Connector pull-out force (see 4.6.5.3). The minimum connector-to-receiver pull-out strength shall be one-half of the specified tensile strength of the attached connector or 100 newtons, whichever is less. There shall be no connector-to-receiver seal damage, distortion or bending of receiver parts, or connector disengagement from the receiver (see 4.6.2.4). The receiver shall meet the requirements of 3.8.2.1.

3.11.4 Fiber dynamic strength (see 4.6.5.4). The minimum fiber dynamic strength shall be 50 percent of the specified proof strength of the attached fiber. There shall be no fiber-to-receiver seal damage or distortion or bending of receiver parts (see 4.6.2.4). The receiver shall meet the requirements of 3.8.2.1.

3.12 Nuclear radiation resistance. Unless otherwise specified (see 3.1), the performance requirements of 3.8.2 and 3.8.3.1 shall be met when tested as specified (see 4.6.6).

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3.13 Electromagnetic effects (see 4.6.7). The receiver shall meet the conducted and radiated emission requirements and the conducted and radiated susceptibility requirements of MIL-STD-461, part 5, as specified (see 3.1). The receiver shall meet the requirements of 3.8.2. Above-deck or below-deck application shall be as specified (see 3.1).

3.14 Electrostatic discharge sensitivity (ESDS) (see 4.6.8). The receiver shall meet the requirements of 3.8, and the requirements for ESDS class 2 devices in accordance with MIL-STD-883.

3.15 Dimensions (see 4.6.9). The receiver shall be enclosed in a physical envelope with dimensions as specified (see 3.1).

3.16 External finish (see 4.6.10). The receiver case finish shall comply with the package finish requirements of MIL-M-38510. The finish shall maintain legibility of the unit part number and required markings (see 3.18). Unless otherwise specified (see 3.1), the pin finish shall comply with MIL-M-38510, finish type B.

3.17 Identification marking (see 4.6.11). Unless otherwise specified (see 3.1), the receiver shall be marked in accordance with MIL-M-38510. The assigned part number shall use the numbering schemes specified (see 6.8).

3.18 Workmanship. Receiver design shall meet all physical and interconnection requirements of this specification. There shall be no damaged parts, loose pins, cracked or abused surfaces, illegible markings, or inconsistencies in finish or appearance. The receiver shall meet the requirements herein for visual inspection prior to final capping of the receiver case and for visual inspection of the finished product (see 4.3.1 and 4.6.2.4).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Screening (see 4.3).
(b) First article inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).

4.3 Screening. Each receiver to be delivered or submitted for inspection shall have been subjected to and passed all of the following screening tests in the order shown (see table II). Receivers with cable pigtails may be subjected to these tests before the pigtail exterior is installed over the coated fiber. For those receivers with pigtails to be cabled after screening, a temporary strain relief may be installed at the receiver/pigtail interface for the screening tests.

TABLE II. Screening inspection.

Inspection	Requirement	Test method
Internal visual	3.18	4.3.1
Burn-in	3.3.1	4.3.2
Random vibration	3.10.8	4.3.3
Temperature cycling	3.10.8	4.3.4
External visual	3.18	4.6.2.4
Hermetic seal	3.5.3	4.6.2.1

4.3.1 Internal visual (see 3.18). Internal visual inspection (pre-cap) shall be performed in accordance with MIL-STD-883, method 2017.

4.3.2 Burn-in (see 3.3.1). Burn-in testing shall be in accordance with MIL-STD-883, method 1015. Unless otherwise specified (see 3.1), all receivers shall be tested for a minimum of 160 hours at the maximum specified temperature (see 3.10.1). Maximum power supply voltage and maximum optical power level shall be applied to the receiver during the test (see 3.8.4, 3.8.1.2 and 3.8.1.3).

4.3.3 Random vibration (see 3.10.8). Random vibration testing shall be in accordance with MIL-STD-883, method 2026, condition II, test condition C. No power shall be applied during this test.

4.3.4 Temperature cycling (see 3.10.8). Temperature cycling shall be in accordance with MIL-STD-883, method 1010, using test condition B. No power shall be applied during this test.

4.4 First article inspection. Inspection shall be as specified in table III and the units submitted for testing shall be representative of the production process. The sample units used for groups II, III, and IV inspections shall have been subjected to and met the requirements of the screening inspections of 4.3 and the group I inspections of table III.

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TABLE III. First article inspection.

Inspection	Requirement	Test method	Sample size
Group I			
Identification marking	3.18	4.6.12	see 4.4.1
Spectral responsivity	3.8.1.1	4.6.3.2.1	1/
Dynamic range	3.8.1.2	4.6.3.2.2	1/
Sensitivity	3.8.1.3	4.6.3.2.3	1/
Signal to noise ratio	3.8.2.1	4.6.3.3.1	1/
Frequency range	3.8.2.2	4.6.3.3.2	1/
Total harmonic distortion	3.8.2.3	4.6.3.3.3	1/
Intermodulation distortion	3.8.2.4	4.6.3.3.4	1/
Amplitude distortion	3.8.2.5	4.6.3.3.5	1/
Group delay	3.8.2.6	4.6.3.3.6	1/
Analog output signal	3.8.3.1	4.6.3.4.1	1/
Short circuit protection	3.8.3.2	4.6.3.4.2	1/
Power supply voltage and current	3.8.4	4.6.3.5	1/
Dimensions	3.15	4.6.9	1/
External finish	3.16	4.6.10	1/
Workmanship	3.18	4.6.2.4	1/
Temperature extremes	3.10.1	4.6.4.1	1/

See footnotes at end of table.

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TABLE III. First article inspection - Continued.

Inspection	Requirement	Test method	Sample size
Group II			
Terminal strength	3.5.4.1	4.6.2.2	1
Solderability	3.5.4.2	4.6.2.3	2/
Thermal shock	3.10.2	4.6.4.2	1
Impact shock	3.10.3	4.6.4.3	3/
Vibration	3.10.4	4.6.4.4	3/
Salt spray	3.10.5	4.6.4.5	3/
Hermetic seal	3.5.3	4.6.2.1	3/
Mechanical tests	3.11	4.6.5	3/
Explosive atmosphere	3.10.6	4.6.4.6	3/
Fungus resistance	3.10.7	4.6.4.7	3/
Group III			
Electromagnetic effects	3.13	4.6.7	1
Electrostatic discharge sensitivity	3.14	4.6.8	4/
Group IV			
Reliability	3.3	4.6.1	see 6.2
Nuclear radiation resistance	3.12	4.6.6	4

1/ The same samples shall be used for all group I inspections.
2/ The same sample shall be used as in the terminal strength inspection.
3/ The same sample shall be used as in the thermal shock inspection.
4/ The same sample shall be used as in the electromagnetic effects inspection.

4.4.1 Samples. Unless otherwise specified (see 6.4), the number of sample units to be subjected to inspection shall be seven receivers, not including those required for reliability testing (see 3.3). For receivers with cable pigtails, those units to be subjected to group II tests may have the pigtail exterior installed after the hermetic seal test. For those receivers with pigtails to be cabled after hermetic seal testing, a temporary strain relief may be installed at the receiver/pigtail interface for those tests prior to hermetic seal testing.

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4.4.2 Failures. One or more failures shall be sufficient cause for refusal to grant first article approval. When first article test units are taken from a larger lot, and the first article is disapproved, none of the units from that lot shall be delivered.

4.5 Quality conformance inspection. Quality conformance inspection shall include the requirements of groups A, B, and C as specified in tables IV, V, and VI (see 6.5). Each receiver shall successfully pass all screening tests (see 4.3) prior to performing group A inspections.

4.5.1 Group A inspection. The group A inspections shall be nondestructive tests, consisting of visual, mechanical, and performance tests as provided in table IV and as specified (see 3.1). Each receiver shall comply with the group A requirements prior to delivery and prior to groups B and C testing, and shall be tested in the order shown.

TABLE IV. Group A inspection.

Inspection	Requirement	Test method
Identification marking	3.18	4.6.12
Spectral responsivity	3.8.1.1	4.6.3.2.1
Dynamic range	3.8.1.2	4.6.3.2.2
Sensitivity	3.8.1.3	4.6.3.2.3
Signal-to-noise ratio	3.8.2.1	4.6.3.3.1
Frequency range	3.8.2.2	4.6.3.3.2
Total harmonic distortion	3.8.2.3	4.6.3.3.3
Intermodulation distortion	3.8.2.4	4.6.3.3.4
Amplitude distortion	3.8.2.5	4.6.3.3.5
Group delay	3.8.2.6	4.6.3.3.6
Analog output signal	3.8.3.1	4.6.3.4.1
Short circuit protection	3.8.3.2	4.6.3.4.2
Power supply voltage and current	3.8.4	4.6.3.5
Temperature extremes	3.10.1	4.6.4.1

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4.5.2 Group B inspection. The group B inspections shall be the nondestructive tests in table V and as specified (see 3.1).

4.5.2.1 Inspection sequence. For receivers with cable pigtails, those units to be subjected to group B inspection may have the pigtail exterior installed after the hermetic seal test. For those receivers with pigtails to be cabled after hermetic seal testing, a temporary strain relief may be installed at the receiver interface for all tests prior to hermetic seal testing. Group B inspections shall be tested in the sequence shown.

TABLE V. Group B inspection.

Inspection	Requirement	Test method
Impact shock	3.10.3	4.6.4.3
Thermal shock	3.10.2	4.6.4.2
Hermetic seal	3.5.3	4.6.2.1
Mechanical tests	3.11	4.6.5

4.5.3 Group C inspection. The group C inspections shall consist of the tests in table VI and as specified (see 3.1). These tests may be destructive to sample parts. Unless otherwise specified (see 6.5), samples for each test shall consist of subsets of group B test samples. For receivers with cable pigtails, the cable exterior may be removed prior to hermetic seal testing in group C inspection.

TABLE VI. Group C inspection.

Inspection	Requirement	Test method
Reliability	3.3	4.6.1
Nuclear radiation resistance	3.12	4.6.6
Terminal strength	3.5.4.1	4.6.2.2
Electromagnetic effects	3.13	4.6.7
Electrostatic discharge sensitivity	3.14	4.6.8
Vibration	3.10.4	4.6.4.4
Hermetic seal	3.5.3	4.6.2.1

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TABLE VI. Group C inspection - Continued.

Inspection	Requirement	Test method
Explosive atmosphere	3.10.6	4.6.4.6
Solderability	3.5.4.2	4.6.2.3
Salt spray	3.10.5	4.6.4.5
Fungus resistance	3.10.7	4.6.4.7

4.6 Test methods.

4.6.1 Reliability (see 3.3). The receiver failure rate shall be determined for class B devices (as defined in MIL-STD-883) using MIL-STD-883, method 1016, except that the minimum ambient temperature shall be 150 degrees Celsius (°C). SNR verification measurements (see 4.6.3.3.1) shall be made using an input optical signal corresponding to the characteristics specified in 3.8.2.2, 3.8.1.2, and 3.8.1.3 at the maximum specified frequency and the maximum optical input power level. The power supply shall be set to the maximum specified magnitude (see 3.8.4). The number of test units shall be as specified (see 6.2).

4.6.2 Design and construction inspections.

4.6.2.1 Hermetic seal (see 3.5.3). The receiver shall be tested in accordance with MIL-STD-883, method 1014, methods A1 and C. For receivers with cable pigtails, see 4.3, 4.4.1, 4.5.2.1, and 4.5.3.

4.6.2.2 Terminal strength (see 3.5.4.1). The receiver electrical terminals shall be tested in accordance with MIL-STD-883, method 2004. Test condition A shall be used for tension; condition B1 shall be used for stress; and condition B2 shall be used for pin fatigue.

4.6.2.3 Solderability (see 3.5.4.2). The receiver electrical pins shall be tested in accordance with MIL-STD-883, method 2003.

4.6.2.4 Visual inspection (see 3.18). Receivers shall be inspected for workmanship in accordance with MIL-STD-883, method 2009.

4.6.3 Performance tests.

4.6.3.1 General. The performance tests for the receiver shall be performed with the receiver having not less than 1 meter of fiber connected. If used, the connector and the fiber or cable comprising the pigtail shall meet the requirements as specified (see 3.1). Measurements shall be made using an optical signal corresponding to the wavelength specified in 1.2.1. The optical signal shall be

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input into the pigtail using the appropriate launch conditions. For multimode fiber, a 70/100 restricted launch shall be used. For single-mode fiber, the pigtail shall be wrapped one turn around a 30-millimeter diameter mandrel.

4.6.3.1.1 Standard test conditions. Unless otherwise specified (see 3.1), the following ambient test conditions shall be used:

(a) Temperature: 25 plus 3, minus 5°C.
(b) Relative humidity: 50 ± 30 percent.
(c) Atmospheric pressure: 725 ± 75 mm of mercury.

4.6.3.2 Optical parameters. Unless otherwise specified (see 3.1), the optical performance parameters shall be verified using the test procedures specified herein and in accordance with the following:

(a) The receiver shall be connected to the apparatus.
(b) The power supply voltage shall be set to the minimum value specified (see 3.8.4).
(c) The optical attenuator shall have sufficient spectral wavelength range, accuracy and resolution to attenuate the optical signal.
(d) A suitable analog input signal with the applicable modulation format and signal amplitude as specified (see 3.1), shall be applied to the input of the analog transmitter.
(e) The analog transmitter shall be complementary to the receiver under test in accordance with the receiver specifications (see 3.1 and 6.1.1).
(f) The results shall be verified in accordance with the requirements of 3.8.1.
(g) The power supply shall be set to the maximum magnitude specified (see 3.8.4). Steps (c) through (f) shall be repeated.
(h) The power supply voltage shall be set to the specified nominal magnitude (see 3.8.4). Steps (c) through (f) shall be repeated.

4.6.3.2.1 Spectral responsivity. As specified in 4.6.3.1, 4.6.3.2, and figure 1, the receiver shall be tested at minimum, maximum, and nominal receiver wavelengths as specified (see 3.8.1.1).

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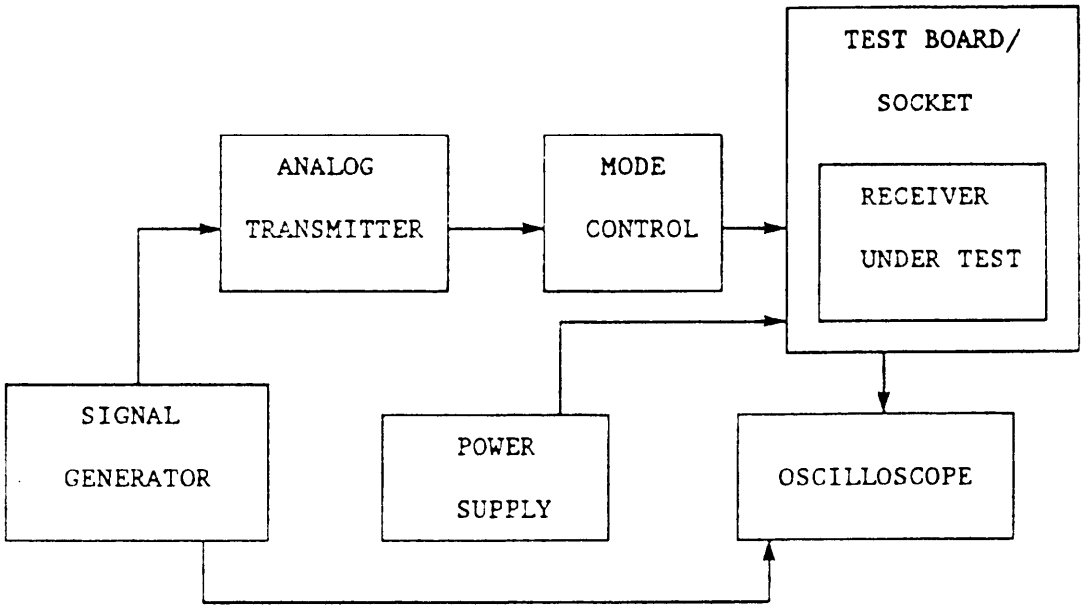


FIGURE 1. Optical performance test set-up.

4.6.3.2.2 Dynamic range (see 3.8.1.2). As specified in 4.6.3.1, 4.6.3.2, and figure 1, the dynamic range shall be measured by applying a sinusoidal input, initializing the set-up to maximum receiver sensitivity, and then decreasing the optical attenuation until the THD exceeds the value specified (see 3.8.2.3).

4.6.3.2.3 Sensitivity. Sensitivity shall be measured in accordance with 4.6.3.1, 4.6.3.2, and figure 1 while maintaining the specified SNR (see 3.8.1.3).

4.6.3.3 Optical-electrical parameters (see 3.8.2). Unless otherwise specified (see 3.1), the optical-electrical performance parameters shall be verified using the test procedures specified in 4.6.3.3.1 through 4.6.3.3.6 and the following:

(a) The receiver shall be connected to the apparatus.
(b) The power supply voltage shall be set to the minimum value specified (see 3.8.4).
(c) The test instrument parameters shall have the capability of measuring the signal under test with a tolerance of 3 percent over the range specified for the device.
(d) A suitable analog input signal with the applicable modulation format and signal amplitude as specified herein, shall be applied to the input of the transmitter.
(e) The analog transmitter used shall be a complement of the receiver under test as specified (see 3.1 and 6.1.1).
(f) The results shall be verified in accordance with the requirements of 3.8.2.
(g) The power supply shall be set to the maximum magnitude specified (see 3.8.4). Steps (c) through (f) shall be repeated.
(h) The power supply voltage shall be set to the specified nominal magnitude specified (see 3.8.4). Steps (c) through (f) shall be repeated.

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4.6.3.3.1 Signal-to-noise ratio (see 3.8.2.1). SNR shall be verified as specified in 4.6.3.1, 4.6.3.3, figure 2 and the following:

(a) The transmitter output shall be linear within plus or minus 1 percent over the frequency test range.
(b) The optical power splitter (coupler) shall have a 50-50 ratio and an excess loss no greater than 0.10 dB.
(c) The SNR of the signal generator shall be at least 10 dB greater than the transmitter and receiver SNR's.
(d) The analyzer shall be set up as follows:

(1) Set the frequency span to zero.
(2) Set the resolution bandwidth to 10 Hz unless otherwise specified (see 3.1).
(3) Set the frequency to the test frequency specified (see 3.1).
(4) An impedance network shall be used at the input to the analyzer, as necessary, so the input impedance of the analyzer will not affect the accuracy of the measurements.

(e) The following procedure shall be performed in sequence:

(1) Disconnect the optical input to the receiver.
(2) Set the generator output to the test frequency specified (see 3.1).
(3) Lower the generator output amplitude to zero.
(4) Adjust the attenuator so that the optical level at the power meter is zero or negligible.
(5) Reconnect the optical input to the receiver.
(6) Disconnect the receiver output to the analyzer.
(7) Terminate the input to the analyzer with the impedance specified for the receiver (see 3.8.3.1)
(8) Measure the maximum noise voltage in dB referenced to 1 milliwatt (dBmW) at the test frequency. Set this value equal to V_{no1}.
(9) Set the reference level of the analyzer to V_{no1}.
(10) Connect the output of the receiver to the terminated input of the analyzer.
(11) Measure the maximum noise voltage in dB with respect to the reference level at the test frequency. Set this value equal to V_{no2}.
(12) Increase the generator output amplitude until the average power measured at the power meter is equal to twice the sensitivity level specified for the receiver (see 3.8.1.3).
(13) Measure the voltage level in dB with respect to the reference level at the test frequency. Set this value to V_{no1}. The output voltage SNR (dB) shall be calculated as follows:

$$SNR\ (dB) = V_{no1} - V_{no2}$$

(f) Steps (d)(3) through (e)(13) shall be performed over the operating frequency range of the receiver at the frequencies specified (see 3.1)

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(g) Steps (d)(3) through (f) shall be performed with the receiver operating in the following ambient temperatures:

(1) 25°C.
(2) Minimum operating temperature of the specified temperature range (see 3.10.1).
(3) Maximum operating temperature of the specified temperature range (see 3.10.1).

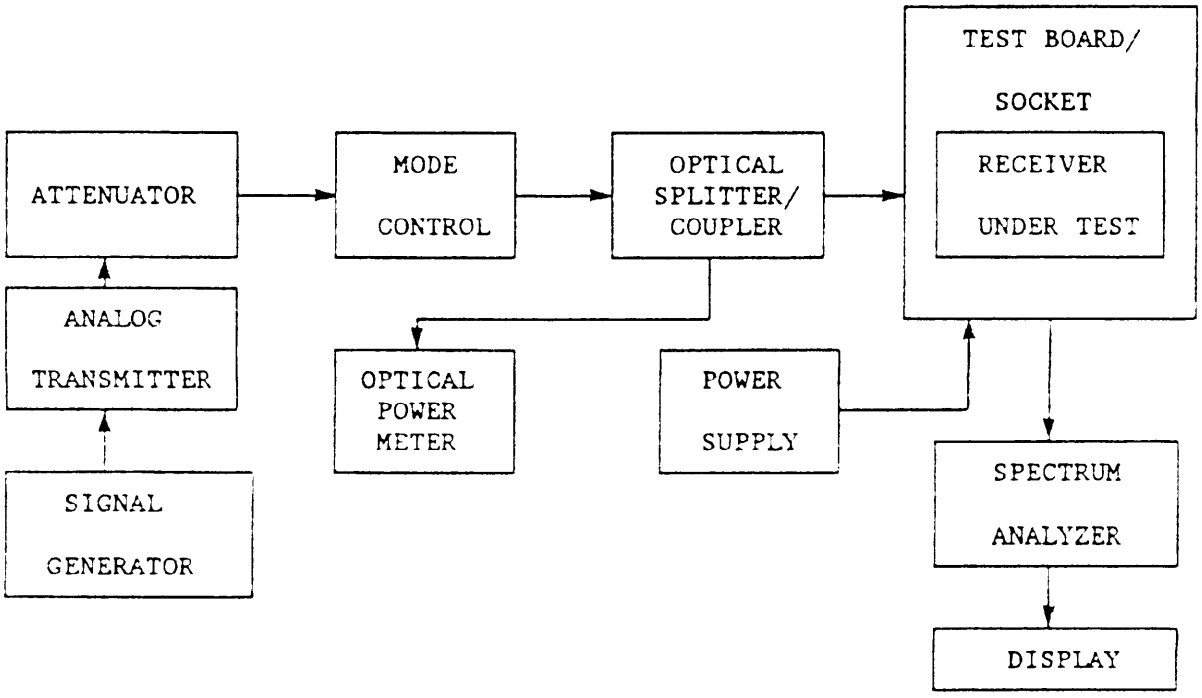


FIGURE 2. SNR test set-up.

4.6.3.3.2 Frequency range. As specified in 4.6.3.1, 4.6.3.3, and the test set-up on figure 3, the signal generator shall inject a signal of constant amplitude into the transmitter. The signal shall be varied over the frequency range specified for the receiver (see 3.8.2.2). The spectrum analysis shall be determined from the recovered signal that shall be displayed at the spectrum analyzer. A transmitter complementary to the receiver under test shall be used.

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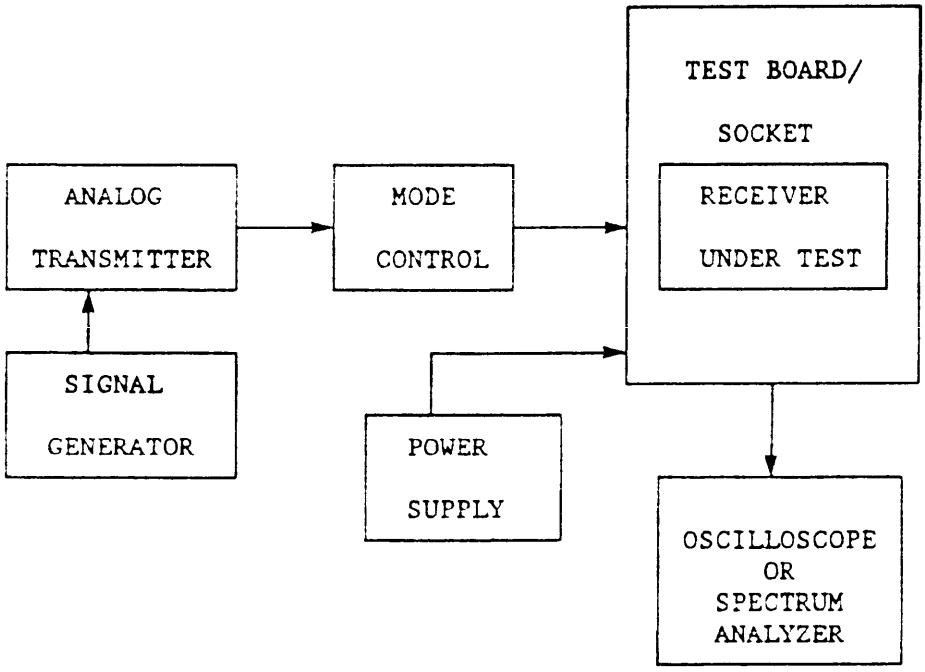


FIGURE 3. Optical-electrical performance test set-up.

4.6.3.3.3 Total harmonic distortion (see 3.8.2.3). As specified in 4.6.3.1, 4.6.3.3, and the test set-up on figure 3, a single frequency signal with a spectral purity better than the THD parameter specified for the receiver (see 3.8.2.3) shall be injected into the complementary transmitter. The THD shall be determined from the spectral analysis obtained from the output of the receiver under test.

(a) The input signal should be of the form

$$x(t) = A_0 \sin \omega_c t$$

(b) For an output of the form

$$y(t) = a_0 + a_1\cos(\omega_0 t + \phi_1) + a_2\cos(2\omega_0 t + \phi_2) + a_3\cos(3\omega_0 t + \phi_3)\ldots,$$

the percent THD shall be equal to

$$\left(\left[\sqrt{\Sigma_{n-2}^{m} a_n^2}\right] / a_1\right) \times 100,$$

where a_n is the peak value of the output at nf_o (where f_o is the frequency associated with the input ω_o). The last frequency to be measured, mf_o, shall be as specified (see 3.1).

(c) The THD shall be measured at the test input frequencies specified (see 3.1) to provide measurement of the THD over the receiver frequency range specified (see 3.8.2.2).

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4.6.3.3.4 Intermodulation distortion (see 3.8.2.3). As specified in 4.6.3.1, 4.6.3.3, and the test set-up shown on figure 3, two constant amplitude single frequency signals shall be injected into the complementary transmitter. The two signals shall be selected in accordance with the following:

(a) The input shall be of the form.

$$x(t) = A \sin \omega_1 t + A \sin \omega_2 t$$

where $A \sin \omega_1 t$ is one of the two input signals and $A \sin \omega_2 t$ is the other input signal. The frequencies f_1 and f_2 associated with ω_1 and ω_2 shall be as specified (see 3.1). Multiple sets of f_1 and f_2 as specified (see 3.1) shall be used for test inputs to provide measurement of the IMD over the receiver frequency range specified (see 3.8.2.2).

(b) Unless otherwise specified, the difference between frequencies f_1 and f_2 shall be no greater than 2 percent of the frequency range specified (see 3.8.2.2).

The percent of IMD shall be measured for any third-order cross products whose frequencies occur within the specified receiver frequency range (see 3.8.2.2) and that are associated with the following:

(a) The output of the receiver shall be of the form

$$y(t) = a_1 x(t) + a_2 x^2(t) + a_3 x^3(t)$$

where $a_1 x(t)$ is the first-order output term and $a_3 x^3(t)$ is the third-order output term.

(b) The amplitudes of the third-order cross products that are part of the third-order output term, to be considered for measurement, shall be at the frequencies $2f_1 + f_2$, $2f_2 + f_1$, $2f_1 - f_2$, and $2f_2 - f_1$.

(c) The percent of distortion at each applicable cross product frequency shall be 100 times the amplitude at the cross product frequency divided by the amplitude of the first order-term a_1.

4.6.3.3.5 Amplitude distortion (see 3.8.2.5). As specified in 4.6.3.1, 4.6.3.3, and the test set-up on figure 3, apply a sinusoidal input signal and sweep the signal source over the frequency range as specified for the receiver (see 3.8.2.2). Obtain a spectral analysis from the output of the receiver under test and determine the amplitude distribution in dB.

4.6.3.3.6 Group delay (see 3.8.2.6). The test set-up shall be as specified in 4.6.3. A narrowband, amplitude modulated, sinusoidal signal, as specified (see 3.1), shall be injected into the complementary transmitter. The sinusoidal signal shall be varied in frequency to cover the operational range. The transmitter shall be coupled to the receiver under test using a pigtail assembly subjected to mode control. An oscilloscope shall be used to display the recovered signal output from the receiver. Group delay shall be measured as the difference in phase of the envelope over the frequency range tested.

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4.6.3.4 Electrical properties (see 3.8.3).

4.6.3.4.1 Analog output signal (see 3.8.3.1). The electrical analog output signal waveform, interface impedance and signal levels shall be verified by the test procedures in 4.6.3.2 and 4.6.3.3.

4.6.3.4.2 Short circuit protection (see 3.8.3.2). The short circuit protection shall be tested in accordance with MIL-STD-883, method 3011.

4.6.3.5 Power supply voltage and current. The maximum current shall be measured and verified in accordance with MIL-STD-883, method 3005, using the nominal supply voltage specified (see 3.8.4).

4.6.4 Environmental tests (see 3.10). The following environmental tests shall be performed as specified in the referenced test methods.

4.6.4.1 Temperature extremes. The performance of the receiver shall be verified as specified in 4.6.3 at both of the operating temperature extremes as specified (see 3.10.1). At each extreme, the receiver shall have attained thermal equilibrium for at least 30 minutes before any measurements are made. The receiver shall be visually inspected for physical damage.

4.6.4.2 Thermal shock (see 3.10.2). The effects of thermal shock on receiver performance shall be tested using MIL-STD-883, method 1011, over the nonoperating temperature range specified (see 3.10.1). Verification of performance shall be conducted in accordance with 4.6.3.4.1 and 4.6.3.3 after the test.

4.6.4.3 Impact shock (see 3.10.3). The receiver shall be subjected to physical shock using MIL-STD-883, method 2002, test condition B. Verification of performance shall be conducted as specified in 4.6.3.3.1 and 4.6.3.4.1 during the test and 4.6.3.3 after the test. The SNR shall be monitored for 5 minutes during which time all five impacts shall be delivered. The SNR during the test shall not decrease more than 1 percent.

4.6.4.4 Vibration (see 3.10.4). The receiver shall be tested for effects of vibration in accordance with MIL-STD-883, method 2026, test condition II, test condition C, and method 2007, test condition A. Verification of performance shall be conducted as specified in 4.6.3.3.1 and 4.6.3.4.1 during the test and 4.6.3.3 after the test.

4.6.4.5 Salt spray (see 3.10.5). The receiver shall be tested for effects of a salt atmosphere in accordance with MIL-STD-883, method 1009, test condition C.

4.6.4.6 Explosive atmosphere (see 3.10.6). The receiver safe operation within explosive atmospheric environments shall be tested in accordance with MIL-STD-202, method 109, under conditions as specified (see 3.1).

4.6.4.7 Fungus resistance (see 3.10.7). Receivers and materials used in the construction of the receivers that are not identified as fungus inert shall be tested in accordance with MIL-STD-810, method 508.

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4.6.5 Mechanical tests (see 3.11).

4.6.5.1 Cable pull-out force (see 3.11.1). The receiver shall be tested in accordance with EIA 455-6. The axial tensile load shall be applied and maintained for 10 minutes. The SNR shall be measured during and after the test (see 4.6.3.3.1).

4.6.5.2 Twist (see 3.11.2). The receiver shall be tested in accordance with EIA 455-36, with the exception that the test sample shall be a properly pigtailed receiver. The number of loads to be applied shall be one. The SNR shall be measured during and after the test (see 4.6.3.3.1).

4.6.5.3 Connector pull-out force (see 3.11.3). The receiver shall be tested in accordance with EIA 455-6, with the exception that the test sample shall be a connector on the receiver. The axial load shall be applied and maintained for 10 minutes. The SNR shall be measured during and after the test (see 4.6.3.3.1).

4.6.5.4 Fiber dynamic strength (see 3.11.4). The fiber pigtail dynamic strength shall be tested as follows: The fiber pigtail shall have an axial tensile load applied at an angle of 45 degrees to the normal (see figure 4). With the load applied, the pigtail shall be rotated through one rotation (360 degrees). The SNR shall be measured during and after the test (see 4.6.3.3.1).

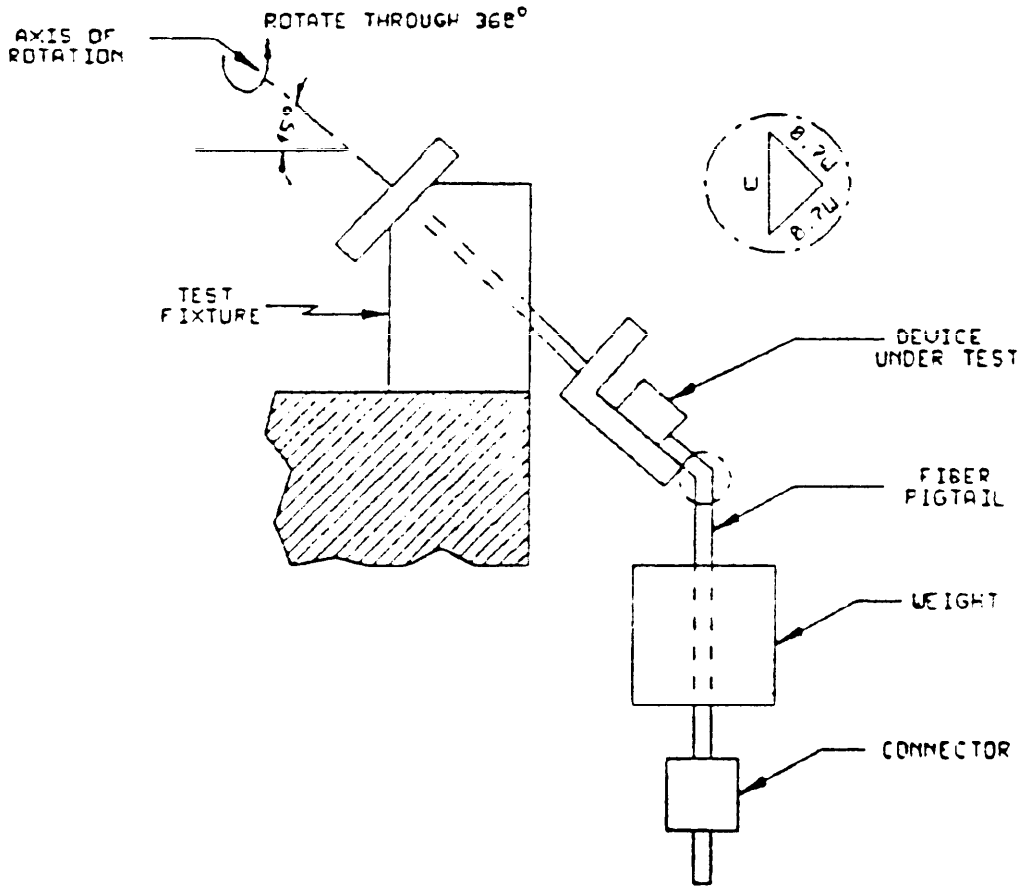


FIGURE 4. Fiber dynamic strength test set up.

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4.6.6 Nuclear radiation resistance (see 3.12). The receiver shall be subjected to radiation in accordance with MIL-STD-883, methods 1017 (neutron) and 1019 (total dose). Verification of performance shall be conducted as specified in 4.6.3.3.1 and 4.6.3.4.1 during the test and 4.6.3.3 after the test. The dose rates and exposure times shall be as specified (see 6.2). The required test temperatures shall be the low and the high operating temperatures and 20°C.

4.6.7 Electromagnetic effects (see 3.13). The receiver shall be tested for effects of electromagnetic emissions in accordance with MIL-STD-462 as applicable. Verification of performance shall be conducted as specified in 4.6.3.4.1 and 4.6.3.3 during and after the test.

4.6.8 Electrostatic discharge sensitivity (see 3.14). The receiver shall be tested for sensitivity to electrostatic discharge in accordance with MIL-M-883, method 3015. The electrostatic pulse shall be not less than 3000 volts or as specified (see 3.1). Verification of performance shall be conducted as specified in 4.6.3 after the test.

4.6.9 Dimensions (see 3.15). The receiver shall be inspected for compliance with unit dimensions and tolerances in accordance with MIL-STD-883, method 2016.

4.6.10 External finish (see 3.16). The external finish of the receiver case shall be verified in accordance with MIL-STD-883, method 2009.

4.6.11 Identification markings (see 3.17). The receiver markings shall be examined for legibility and required information in accordance with MIL-STD-883, method 2009. The marking shall be legible and complete, and shall be resistant to solvents in accordance with MIL-STD-883, method 2015.

4.6.12 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified herein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.7.)

5.1 Preservation and packing. The receiver shall be preserved and packed for shipment in accordance with MIL-M-55565.

5.2 Marking. Marking on the receiver packaging shall be in accordance with MIL-M-38510.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. The receivers specified herein are intended for the reception of optically transmitted analog signals. The receiver applications may utilize any signal waveform within the performance capabilities as specified herein.

6.1.1 Engineering information. MIL-T-24735 is the companion specification for analog fiber optic shipboard transmitters. Although the companion specification is not involved in the classification or testing of receivers, it is often useful for evaluating the fiber optic link system. A single-mode receiver will use a multimode pigtail.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of the specification.
(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(c) Applicable specification sheet number, title, and date.
(d) Whether first article inspection is required (see 3.2).
(e) Reliability testing is required, unless waived by the contracting officer. The number of test units must be specified (see 3.3 and table III).
(f) The number of test units needed (see 4.6.1).
(g) Nuclear radiation dose rates and exposure times must be specified when this test is required (see 4.6.6).
(h) Part number of receiver (see 6.8).
(i) Quantity required.

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.4	DI-E-2121	Certificate of compliance	----

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the items should be a first article sample. The first article should consist of 7 units. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test

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results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4.1 Lot size. The lot size may be specified contractually, if desired.

6.5 Quality conformance. Quality conformance inspections require contractual definition of the overall test program, including sample sizes and lot sizes, if appropriate (see 4.5).

6.6 Definitions.

6.6.1 Amplitude distortion. Amplitude distortion is distortion of a signal in a system, subsystem, or device when the output amplitude is not a linear function of the input amplitude under specified operating conditions.

6.6.2 Analog signal. An analog signal is a nominally continuous signal that is considered to be continuously variable and whose magnitude is made directly proportional to the data or to a suitable function of the data.

6.6.3 Cladding mode stripper. A cladding mode stripper is a material applied to cladding to allow electromagnetic energy being transmitted in the cladding to leave the cladding.

6.6.4 Dynamic range. Dynamic range is the difference in decibels between the noise level or minimum acceptable signal level and its over load level.

6.6.5 Fiber cladding. Fiber cladding is a transparent material that surrounds the core of an optical fiber and that has a lower index of refraction than the core material.

6.6.6 Fiber core. A fiber core is the central portion of an optical fiber. The core has a higher refractive index than the cladding that surrounds it. The bulk of light energy is confined to and propagates in the core. Single-mode fiber core diameters range from 2 to 11 μm, depending on the numerical aperture, core radius, and wavelength of the incident light.

6.6.7 Frequency range. Frequency range is the difference between the limiting frequencies of a continuous frequency band.

6.6.8 Group delay. Group delay is the distortion of a waveform made up of two or more different frequencies, caused by the difference in arrival time of each frequency at the output of a transmission system.

6.6.9 Interface. An interface is a concept involving the specification of the interconnection between two equipments of systems. The specification includes the type, quantity, and function of the interconnection circuits, and the type and form of signals to be interchanged by means of those circuits.

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6.6.10 Intermodulation distortion. Intermodulation distortion is a nonlinear distortion characterized by the appearance of frequencies in the output equal to the sums and differences of integral multiples of the component frequencies present in the input.

6.6.11 Optical cable pigtail. An optical cable pigtail is a short length of fiber optic cable permanently fixed to a component and is used to couple optical power between it and another optical cable pigtail or fiber optic cable, such as a transmission cable.

6.6.12 Optical fiber pigtail. An optical fiber pigtail is a short length of optical fiber permanently fixed to a component and is used to couple optical power between it and another optical fiber pigtail or a transmission fiber.

6.6.13 Peak emission wavelength. The peak emission wavelength is the wavelength at which a light source emits its maximum optical power.

6.6.14 Sensitivity. Sensitivity is the minimum input signal required by a device to produce a specified output signal having a specified SNR. The sensitivity may be expressed as power in dBm or as field strength in microvolts/meter, with input network impedance stipulated.

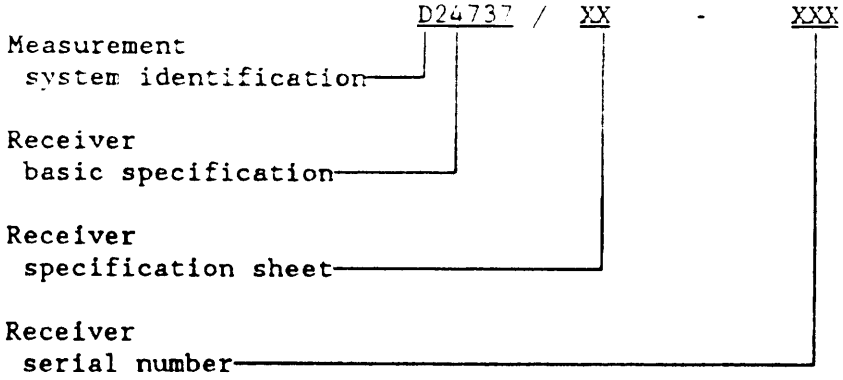
6.6.15 Signal-to-noise ratio. SNR is the ratio of the amplitude of the desired signal to the amplitude of noise signals.

6.6.16 Total harmonic distortion. THD is the ratio of the sum of the powers of all harmonic frequency signals (other than the fundamental) to the power of the fundamental frequency signal at the output. This ratio is measured under specified conditions and is expressed in decibels.

6.7 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.8 Part number.

6.8.1 Part or Identifying Number (PIN). The PIN's to be used for receivers acquired to this specification are created as follows:



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Examples: D24737/01-001
D24737/01-002

6.9 Part designator. A part designator if specified (see 3.1) should include classifications (see 1.2) as follows:

(a) Wavelength class (see 1.2.1).
(b) Frequency range (see 1.2.2).
(c) Style (see 1.2.3).

Example: A2A

6.10 Specification use. Where MIL-C-85045 is referenced, the Navy version, MIL-C-0085045, should be used until cancelled.

6.11 Subject term (key word) listing.

Avalanche photo diode
Fiber optic communications
Fiber optic component
Multimode receiver
Optical amplifier
Optical connectorized component
Optical detector
Optical pigtailed component
Photodiode
Single mode receiver
Surface mount component

Review activities:
Navy - EC, YD

Preparing activity:
Navy - SH
(Project 6026-N002)

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-24737(NAVY)	RECEIVERS, LIGHT SIGNAL, ANALOG, FIBER OPTIC, SHIPBOARD (METRIC), GENERAL SPECIFICATION FOR

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
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	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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